

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Task 1

TP=50, TN=35, FP=5, FN=10

$$\begin{aligned} Accuracy &= (TP+TN)/(TP+TN+FP+FN) \\ &= (50+35)/(50+35+5+10) \\ &= 85/100 = 0.85 \end{aligned}$$

$$\begin{aligned} Precision &= TP/(TP+FP) \\ &= 50/(50+5) \\ &= 50/55 = 0.91 \end{aligned}$$

$$\begin{aligned} Recall &= TP/(TP+FN) \\ &= 50/(50+10) \\ &= 50/60 = 0.83 \end{aligned}$$

$$\begin{aligned} F1\text{-score} &= 2*(Precision*Recall)/(Precision+Recall) \\ &= 2*(0.91*0.83)/(0.91+0.83) \\ &= 0.87 \end{aligned}$$

Task 2

TP=70, TN=95, FP=15, FN=20

$$\text{Accuracy} = (70+95)/(70+95+15+20) \\ = 165/200 = 0.825$$

$$\text{Precision} = 70/(70+15) \\ = 70/85 = 0.82$$

$$\text{Recall} = 70/(70+20) \\ = 70/90 = 0.78$$

$$\text{F1-score} = 0.80$$

Task 3

$$\text{TP}=120, \text{TN}=125, \text{FP}=25, \text{FN}=30$$

$$\text{Accuracy} = 245/300 = 0.817$$

$$\text{Precision} = 120/(120+25) = 0.83$$

$$\text{Recall} = 120/(120+30) = 0.80$$

$$\text{F1-score} = 0.81$$

Task 4

$$\text{TP}=45, \text{TN}=70, \text{FP}=20, \text{FN}=15$$

$$\text{Accuracy} = 115/150 = 0.767$$

$$\text{Precision} = 45/(45+20) = 0.69$$

$$\text{Recall} = 45/(45+15) = 0.75$$

$$\text{F1-score} = 0.72$$

Task 5 (Multi-class A,B,C)

$$\text{Accuracy} = (40+35+40)/150 = 0.77$$

Class A:

$$\text{Precision} = 40/(40+9)=0.82$$

$$\text{Recall} = 40/(40+10)=0.80$$

$$\text{F1} = 0.81$$

Class B:

$$\text{Precision} = 35/(35+12)=0.74$$

$$\text{Recall} = 35/(35+15)=0.70$$

$$\text{F1} = 0.72$$

Class C:

$$\text{Precision} = 40/(40+14)=0.74$$

$$\text{Recall} = 40/(40+10)=0.80$$

$$\text{F1} = 0.77$$

Task 6 (Cat, Dog, Rabbit)

$$\text{Accuracy} = (50+45+41)/170 = 0.80$$

Class: Cat

$$\text{TP} = 50$$

$$\text{FP} = 10 + 3 = 13$$

$$\text{FN} = 8 + 2 = 10$$

Precision

$$= \text{TP} / (\text{TP} + \text{FP})$$

$$= 50 / (50 + 13)$$

$$= 50 / 63$$

$$= \mathbf{0.79}$$

Recall

$$= \text{TP} / (\text{TP} + \text{FN})$$

$$= 50 / (50 + 10)$$

$$= 50 / 60$$

$$= \mathbf{0.83}$$

F1-score

$$= 2 \times (0.79 \times 0.83) / (0.79 + 0.83)$$

$$= 2 \times (0.6557) / 1.62$$

$$= \mathbf{0.81}$$

Class: Dog

$$\text{TP} = 45$$

$$\text{FP} = 8 + 6 = 14$$

$$FN = 10 + 5 = 15$$

Precision

$$= 45 / (45 + 14)$$

$$= 45 / 59$$

$$= \mathbf{0.76}$$

Recall

$$= 45 / (45 + 15)$$

$$= 45 / 60$$

$$= \mathbf{0.75}$$

F1-score

$$= 2 \times (0.76 \times 0.75) / (0.76 + 0.75)$$

$$= 2 \times (0.57) / 1.51$$

$$= \mathbf{0.75}$$

Class: Rabbit

$$TP = 41$$

$$FP = 2 + 5 = 7$$

$$FN = 3 + 6 = 9$$

Precision

$$= 41 / (41 + 7)$$

$$= 41 / 48$$

$$= \mathbf{0.85}$$

Recall

$$= 41 / (41 + 9)$$

$$= 41 / 50$$

$$= \mathbf{0.82}$$

F1-score

$$= 2 \times (0.85 \times 0.82) / (0.85 + 0.82)$$

$$= 2 \times (0.697) / 1.67$$

$$= \mathbf{0.83}$$

Task 7 (A,B,C,D)

$$\text{Accuracy} = (30+28+32+36)/200 = 0.63$$

Class A

$$\text{TP} = 30$$

$$\text{FP} = 4 + 3 + 2 = 9$$

$$\text{FN} = 5 + 3 + 2 = 10$$

Precision

$$= 30 / (30 + 9)$$

$$= 30 / 39$$

$$= \mathbf{0.77}$$

Recall

$$= 30 / (30 + 10)$$

$$= 30 / 40$$

$$= \mathbf{0.75}$$

F1-score

$$= 2 \times (0.77 \times 0.75) / (0.77 + 0.75)$$

$$= \mathbf{0.76}$$

Class B

$$\text{TP} = 28$$

$$\text{FP} = 5 + 7 + 3 = 15$$

$$\text{FN} = 4 + 6 + 2 = 12$$

Precision

$$= 28 / (28 + 15)$$

$$= 28 / 43$$

$$= \mathbf{0.65}$$

Recall

$$= 28 / (28 + 12)$$

$$= 28 / 40$$

$$= \mathbf{0.70}$$

F1-score

$$= \mathbf{0.67}$$

Class C

$$TP = 32$$

$$FP = 3 + 6 + 4 = 13$$

$$FN = 3 + 7 + 3 = 13$$

Precision

$$= 32 / 45 = \mathbf{0.71}$$

Recall

$$= 32 / 45 = \mathbf{0.71}$$

F1-score

$$= \mathbf{0.71}$$

Class C

$$TP = 32$$

$$FP = 3 + 6 + 4 = 13$$

$$FN = 3 + 7 + 3 = 13$$

Precision

$$= 32 / 45 = 0.71$$

Recall

$$= 32 / 45 = 0.71$$

F1-score

$$= \mathbf{0.71}$$